



Protective effect of *Berberis aristata* against peritonitis induced by carbapenem-resistant *Escherichia coli* in a mammalian model



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ABSTRACT

Objectives: *Berberis aristata* is known to contain a variety of phenolic compounds contributing to its holistic capability of mitigating bacterial multidrug resistance.

Methods: *B. aristata* stem bark extract was prepared and was characterised using liquid chromatography–mass spectrometry (LC–MS). The antimicrobial efficacy of the extract against carbapenem-resistant *Escherichia coli* was assessed in vivo in an animal model using Sprague Dawley rats. Microbial counts in blood and urine, physical health status, haematological and biochemical analysis of blood, and histopathology of the kidney were assessed as the study endpoints.

Results: An aquo-alcoholic extract of *B. aristata* (PTRC-2111-A) was found to effectively manage peritonitis induced by carbapenem-resistant *E. coli* in a rat model at a single post-exposure prophylactic dose of 0.5 mg/kg body weight (BW). The extract was also found to show a no observed adverse effect level (NOAEL) up to a dose of 2000 mg/kg BW. Physical, immunological, haematological, biochemical and histopathological aberrations were found to be restored to normal in the herbal-treated group at a dose of 0.5 mg/kg BW.

Conclusions: The antimicrobial and hepatorenal protective ability of PTRC-2111-A could be attributed to the presence of isoquinoline alkaloids.

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1. Introduction

The evolution of Gram-negative commensal bacteria into resistant forms harbouring genes encoding carbapenemases is an emerging threat. Prominent antibiotic resistance mechanisms such as *Klebsiella pneumoniae* carbapenemase (KPC), New Delhi metallo- β -lactamase-1 (NDM-1) etc. might lead to an antibiotic apocalypse heading into a post-antibiotic era [1]. The devastating impact of such extensive drug resistance includes urinary tract, bloodstream and intra-abdominal infections, which are difficult to manage and thus a major future clinical challenge [2,3]. The prevalence of carbapenem-resistant Enterobacteriaceae (CRE) in India has been reported to vary from 7.87% to 51%, whereas the global prevalence has been estimated at 5–50% [4]. Moreover, death rates attributed to nosocomial infections with

carbapenemase-producers are high (>50%), with 18–67% mortality associated with metallo- β -lactamase-producers only [5,6]. NDM is the most widespread carbapenemase with an intricate pattern of horizontal antibiotic resistance gene transfer [7,8]. The main cause of such disseminated infections could be either autochthonous human carriage or the external environment [9]. Available therapeutic regimens for the treatment of such multidrug-resistant infections are limited. Also, the clinical complications and contraindications, i.e. organ toxicity, hypersensitivity etc., associated with the available chemotherapeutics call for an alternative medicinal approach employing holistic herbs and natural plant products [10].

In our previous studies, a novel herbal informatics approach was used to identify potent natural plant products against carbapenem-resistant *Escherichia coli* by employing a matrix-based analysis of a database of more than 40 traditional medicinal plants [11]. Further in vitro studies suggested *Berberis aristata* as one of the potent leads, having an important minimum inhibitory concentration (MIC) (ca. 25 μ g/mL).

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B. aristata is a well-known traditional medicine owing to its antimicrobial, anti-inflammatory, antipyretic and antiseptic properties [12,13]. In addition, it is reported that the root and stem bark of ethanolic extracts of *B. aristata* have antimicrobial activity in vitro against *Vibrio cholerae*, *Staphylococcus* spp., *Candida* spp. and *Aspergillus* spp. etc. [12,14]. However, to the best of our knowledge, to date no studies regarding the in vivo antimicrobial activity of *B. aristata* against carbapenem-resistant *E. coli* have been conducted. Therefore, the main aim of this study was to evaluate the antimicrobial activity of an aquo-ethanolic extract of *B. aristata* stem bark using a mammalian model system.

2. Materials and methods

2.1. Isolation and identification of the test organism

Clinical samples of biological fluids were collected from different regional pathology laboratories of the National Centre for Disease Control (New Delhi, India). Collected samples were immediately enriched in sterile nutrient broth at 37 °C for 48 h. Following incubation, a loopful of culture was plated on MacConkey agar. Carbapenem-resistant *E. coli* isolates were identified on the basis of morphological, cultural and biochemical characteristics as described previously [15]. A loopful of culture was inoculated into 5 mL of fresh sterile nutrient broth and was incubated at 37 °C for 6–8 h. The turbidity of the nutrient broth was adjusted to a 0.5 McFarland standard (ca. 1×10^8 CFU/mL) by standard procedure [16] and this suspension was used as the inoculum. All rats were inoculated with the same isolate of *E. coli*.

2.2. Herbal extract preparation

Stem bark of *B. aristata* was shade dried, was mechanically powdered and was exhaustively extracted using 1:1 water:ethanol continuously for 4–5 h at a temperature of ca. 55 °C, followed by continuous condensation and collection of the marc [17,18]. The residual solvent of the concentrated extract was evaporated under vacuum in a rotary evaporator (Cole-Parmer, Mumbai, India) and was spray-dried to develop an amorphous powder. The extract was designated as PTRC-2111-A [17].

2.3. Liquid chromatography–mass spectrometry (LC–MS) analysis

Herbal extract PTRC-2111-A, dissolved in a mixture of ethanol and water (1:1 v/v), was analysed on a Shimadzu LC-6A HPLC machine (Shimadzu, Kyoto, Japan) utilising a Waters Nova-Pak C18 column (250 × 4.6 mm, 4 µm) (Waters, Delhi, India) with an auto-injector (SIL-10ADvp; Agilent Technologies, Delhi, India) and diode array detector (SPD-M10A; Gilson Inc., Middleton, WI). Elution was carried out at a flow rate of 1.0 mL/min under a linear gradient of acetonitrile in 0.2% trifluoroacetic acid (solvent A) and water with 0.2% formic acid (solvent B) from 5% to 100% A. The equilibration time was 50 min with a four-step gradient of geometric progression. LC–MS analysis was performed using a LCQ MS mass spectrometer (Thermo Fisher, New Delhi, India) [19]. Comparative analysis of the eluted peak data with respect to mass spectral data of reference compounds was used to identify known compounds in PTRC-2111-A.

2.4. Animals and conditions

Four-week-old male Sprague Dawley rats, weighing 130–150 g, bred locally in the animal house of the Institute of Nuclear Medicine and Allied Sciences (INMAS) (New Delhi, India), were used for this experiment. Rats were acclimatised for 7 days before starting the experiments. During the acclimatisation and

experimental periods, rats were housed in polycarbonate cages in a room maintained under standard laboratory conditions [Institutional Animal Ethics Committee of INMAS; Indian National Science Academy (INSA)] [20].

2.5. Assessment of the maximum tolerable dose (MTD)

Various doses of PTRC-2111-A ranging from 0.25–2000 mg/kg body weight (BW) were given orally to a group of uninoculated Sprague Dawley rats ($n=6$). Each rat was assessed for a period of 5 days. Any lethality, if observed, was recorded and consequently the dose above which any abnormal symptoms/death was observed would be considered as the MTD [21].

2.6. Carbapenem-resistant Enterobacteriaceae-induced peritonitis infection model

The peritonitis infection model was standardised with respect to route of inoculation, i.e. oral, intraperitoneal or intravenous, and respective colony counts were monitored in blood and urine daily up to a period of 5 days, necessary for inducing systemic infection. The time at which the microbial count reached its maximum peak was designated as the time of administration of herbal extract. The route of administration of PTRC-2111-A (1:1 ethanol:water), i.e. intravenous, intraperitoneal or oral, was standardised with respect to efficacy analysis in terms of ability to reduce the microbial count in blood and urine compared with the respective controls [22].

2.7. In vivo analysis of activity against carbapenem-resistant Enterobacteriaceae

Sprague Dawley rats were distributed into nine groups ($n=6$). Herbal treatment groups (herbal+infection) included doses of 0.25–10 mg/kg BW with selected inoculation and drug administration routes (as standardised in the previous step). Furthermore, one uninfected group (herb-only group; no *E. coli* challenge) received herbal extract (10 mg/kg). The test groups were compared with the negative control (infection only, administered with 1 mL of *E. coli* bacterial suspension, ca. 1×10^8 CFU/mL) and the uninoculated control group (administered with water only). Efficacy analysis of PTRC-2111-A involved enumeration of *E. coli* colonies in blood and urine samples of each animal on the first, third and fifth day of the disease cycle [23,24]. The killing index was also calculated using the formula: killing index = (probable colonies–survival colonies)/(probable colonies). Probable colonies are presumably the number of colonies counted on plates that were inoculated with blood from rats that were inoculated with bacteria but were not treated with herbal extract. Furthermore, rats were also observed for visible symptoms, if any, for a period of 5 days.

2.8. Analysis of physical health status

The 5-day physical examination was performed based on previous studies [25]. Animals from each of the described groups were examined for any significant physical changes, i.e. signs of morbidity and frequent urination. Physical conditions in terms of body weight and temperature were recorded for each animal on alternate days during the 5-day study period.

2.9. Haematology and biochemistry

The test groups were analysed for their haematological status using standard biochemical assays. The test groups were compared with the negative control and uninoculated control groups, respectively [26].

2.9.1. Whole blood analysis

Blood samples were analysed for complete blood profile including white blood cell count, red blood cell count, haemoglobin, haematocrit/packed cell volume (Hct), mean corpuscular volume (MCV), mean corpuscular haemoglobin (MCH), mean corpuscular haemoglobin concentration (MCHC), platelet count (PLT), lymphocytes, monocytes, eosinophils, granulocytes, red blood cell distribution width (RDW), mean platelet volume (MPV) and platelet distribution width (PDW). Measurements were performed using a Hematology Analyzer (Celltac Nihon IN530; Nihon Kohden, Tokyo, Japan). The standard kits used for the respective analysis were obtained from Randox Laboratories Ltd. (Mumbai, India).

2.9.2. Serum-based analysis

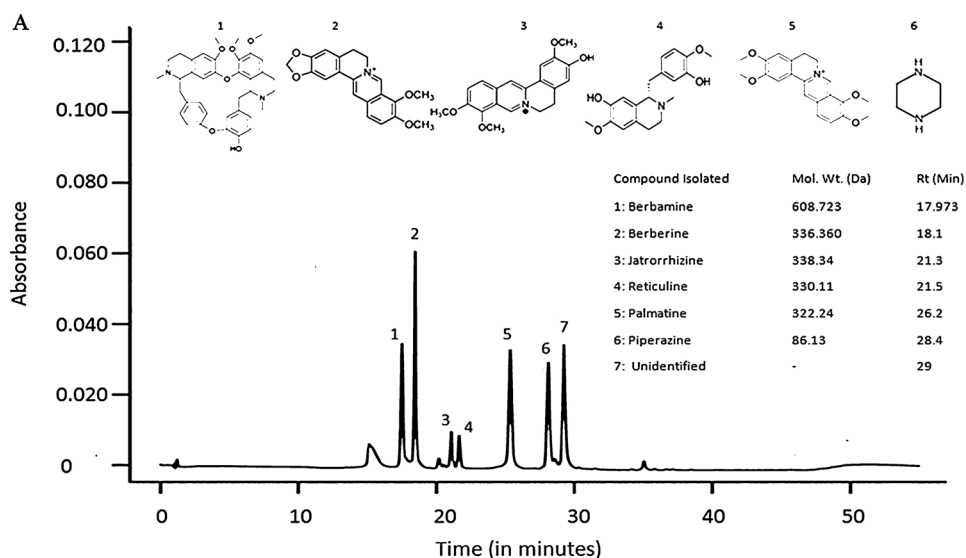
Serum aliquots were subjected to evaluation of alanine aminotransferase (ALT), aspartate aminotransferase (AST), lactate dehydrogenase (LDH), γ -glutamyl transferase (GGT) and alkaline phosphatase (ALP) activities, albumin/globulin (A/G) ratio, urea, creatinine and total antioxidant status (TAS) using a biochemistry analyser (Selectra E; Vital Diagnostics, Smithfield, Rhode Island). The reagent kits used for the tests were obtained from Randox Laboratories Ltd.

2.10. Histopathological assessment

Renal tissue samples of sacrificed Sprague Dawley rats from the most effective dosage group (herbal + infection), herb-only group (effective dose), negative control (infection only) and uninoculated (water only) control groups were carefully removed on Day 5 of experimentation. Samples were fixed in 10% formalin, were dehydrated in graded alcohol, were embedded in paraffin and were cut into 4–5 μ m thick sections as per standard procedures [27]. Haematoxylin–eosin was used to stain the sections followed by photomicroscopic assessment using a Model N-400ME photomicroscope (CEL-TECH Diagnostics, Hyderabad, India).

2.11. Statistical analysis

All data were obtained from three independent experiments with each group of six animals. Data were analysed using IBM SPSS Statistics v.22.0 (IBM Corp., Armonk, NY). Statistical differences were assessed by one-way analysis of variance (ANOVA) followed by post-hoc analysis. *P*-values of <0.05 were considered statistically significant.



B Phytocompounds identified using LC/MS profiling and LC/MS sub-structural Analysis of PTRC-2111-A

Sample	Compound(s)	Retention Time (Min)	Molecular weight	Other fragments peak
Std 1	Berbamine	17.30	608.443	360.03, 314.96
Std 2	Berberine	18.60	336.4	282.92, 88.92
Std 3	Jatrorrhizine	21.20	338.27	312.93, 98.32, 66.56
Std 4	Reticuline	21.10	330.09	324.94, 306.00
Std 5	Palmatine	26.40	322.43	312.95, 245.89
Std 6	Piperazine	28.10	86.36	84.1, 78.2, 73.7, 72.79, 69.90, 59.3, 46.02
PTRC-2111-A	Berbamine	17.97	608.723	360, 364.46
	Berberine	18.10	336.360	334.12, 330.98
	Jatrorrhizine	21.30	338.340	336.12, 330.1
	Reticuline	21.50	330.110	324.00, 305.50
	Palmatine	26.20	322.240	322.05, 320.45
	Piperazine	28.40	86.130	82.1, 85.2, 86.03
	Unidentified compound	29.00	-	-

Fig. 1. Liquid chromatography–mass spectrometry (LC–MS) profile of aquo-alcoholic extract of the stem bark of *Berberis aristata* (PTRC-2111-A). (A) High-performance liquid chromatography (HPLC) chromatogram of the six major isoquinoline alkaloids of PTRC-2111-A detected at 275 nm. Peaks were numbered in order of their elution. (B) Chromatographic and mass spectroscopic characteristics of the major identified isoquinoline alkaloids in PTRC-2111-A. One of the peaks was unidentified [retention time (RT) of ca. 29 min].

3. Results

3.1. Liquid chromatography–mass spectrometry analysis

LC–MS fingerprint analysis of the aquo-alcoholic extract PTRC-2111-A revealed seven peaks at retention times (RTs) of 17.973, 18.1, 21.3, 21.5, 26.2, 28.4 and 29 min ($\lambda = 275$ nm) (Fig. 1A). The eluted samples were subjected to mass spectral analysis and the fragmentation patterns of major fragment ions were recorded. The fragmentation pattern, as revealed in full ion spray mass spectrum, illustrated 15 distinct peaks, which were compared with a reference molecular weight-linked database. Accordingly, six compounds were identified, namely berbamine (RT = 17.973 min), berberine (RT = 18.1 min), jatrorrhizine (RT = 21.3 min), reticuline (RT = 21.5 min), palmatine (RT = 26.2 min) and piperazine (RT = 28.4 min) (Fig. 1B).

3.2. Assessment of the maximum tolerable dose (MTD)

MTD analysis of PTRC-2111-A in the dose range of 0.25–2000 mg/kg BW (oral) exhibited no signs of any acute toxicity up to the maximum administered dose of 2000 mg/kg BW. There were no significant changes in body weight, body temperature or general activity up to a post-administration period of 5 days, indicating a no observed adverse effect level (NOAEL).

3.3. Carbapenem-resistant *Enterobacteriaceae*-induced peritonitis infection model

First, the route of inoculation of the CRE bacterial suspension was standardised by evaluating the number of colonies observed in blood and urine samples after 24 h of inoculation. The number of colonies achieved following inoculation was found to be significantly ($P < 0.05$) higher by the intraperitoneal route, with an observed colony count of 79 ± 0.94 compared with colony counts of 69 ± 0.95 and 18 ± 0.95 for the intravenous and oral routes, respectively. This implied that intraperitoneal administration was the most preferable route for inducing systemic infection (Fig. 2A). The maximum number of colonies was observed after 24 h of inoculation and hence the peak infection period was optimised as 24 h (Day 1) in accordance with the infection cycle of the micro-organism within the murine system. Accordingly, the time for post-exposure prophylaxis was finalised as 24 h post-inoculation (Day 1) (Fig. 2B). The route of herbal drug administration was optimised as the oral route in view of the fact that maximum efficacy was observed with the oral route compared with the other routes (Fig. 2C). Blood and urine sampling was performed on Days 2, 4 and 6.

3.4. In vivo analysis of activity against carbapenem-resistant *Enterobacteriaceae*

Analysis of the anti-CRE potential was evaluated for a dose range of 0.25–10 mg/kg BW (oral) with relative reduction in microbial counts in blood and urine samples. The maximum reduction in microbial count (anti-CRE potential, ca. $99.3 \pm 0.5\%$ at Day 1 after administration of herbal extract) was observed at 0.5 mg/kg BW (oral) both in blood and urine samples. Multivariate ANOVA was conducted to compare the effects of time and sample type (urine or blood) on microbial count in the negative (infection only) and uninoculated (water only) control groups, the herb-only group and the treated groups. There was a significant variation in mean microbial count for all treatment groups [$F(2, 144) = 2355.319, P = 0.001$]. The mean $\pm$ standard deviation (S.D.) microbial count for the infected group (191.5 ± 2.25) was significantly higher than that for the negative control group

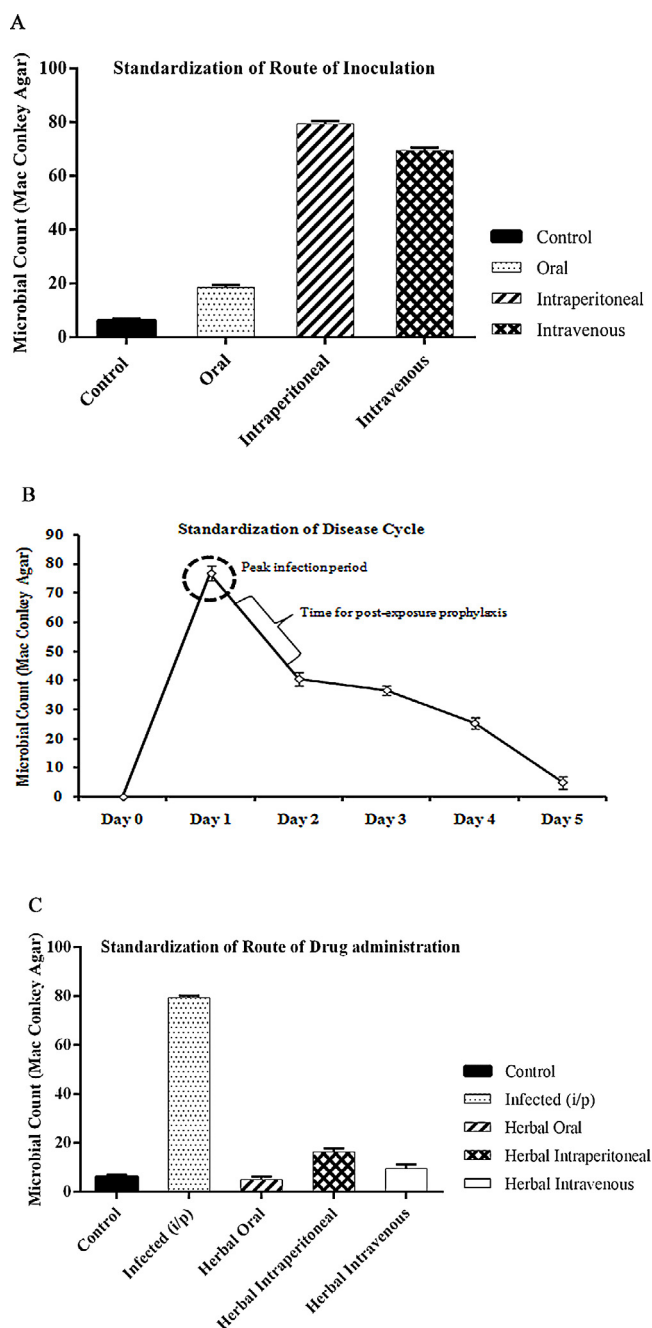


Fig. 2. Standardisation of the infection cycle in the peritonitis model. (A) Standardisation of the route of inoculation of the carbapenem-resistant *Enterobacteriaceae* (CRE) bacterial suspension (10^8 CFU/mL). The number of CRE colonies post-inoculation was found to be significantly ($P < 0.05$) higher by the intraperitoneal route (79 ± 0.94) compared with the intravenous (69 ± 0.95) and oral (18 ± 0.95) routes. (B) Standardisation of the disease cycle of CRE. The peak infection period was optimised as 24 h (Day 1), as the microbial count was recorded to be highest after 24 h. Accordingly, the time for post-exposure prophylaxis was finalised as 24 h post-inoculation. (C) Standardisation of the route of drug administration. The route of PTRC-2111-A administration was optimised as the oral route, which showed maximum efficacy (ca. 97% reduction in microbial count) compared with the other routes, i.e. intraperitoneal (ca. 80% reduction in microbial count) and intravenous (ca. 88% reduction in microbial count).

(6.17 ± 0.51). However, the mean microbial load was significantly reduced in the treatment group at a dose of 0.5 mg/kg BW (18.23 ± 2.27). Furthermore, the herb-only group (6.16 ± 0.94) did not significantly differ from the negative control group. Similarly, there was significant variation in mean microbial count with

respect to biological samples (urine and blood) for all treatment groups [$F(1, 144) = 23\ 508.178$, $P = 0.001$]. The mean $\pm$ S.D. microbial count in urine (109.69 ± 1.02) was significantly higher than the mean microbial count in blood (55.84 ± 6.29) for all tested groups. Furthermore, over the study period, at the lowest treatment dosage of 0.25 mg/kg a progressive increase in killing index was observed from 56% on Day 1 to 81% on Day 5 in the case of blood samples and from 22.6% on Day 1 to 48% on Day 5 in the case of urine samples. However, beyond 0.25 mg/kg BW, a killing index of nearly 100% was observed at all dosages. These results clearly suggested that 0.25 mg/kg BW can be assumed as the lowest dose at which bacteriostatic interactions begin, however 0.5 mg/kg BW remains as the therapeutic dose at which bactericidal action begins (Fig. 3).

Furthermore, the untreated infected group exhibited a killing index almost less than 50%, suggestive of the basal immunological clearance of bacterial load in the murine system; however, this clearance was not absolute, whereas in the treatment groups the

microbial load was significantly reduced, indicating the extensive antibacterial action of the compound.

3.5. Analysis of physical health status

There was a significant ($P < 0.05$) decrease in the mean $\pm$ S.D. weight of all infected groups (126.83 ± 2.48 g) compared with the negative control group (148.43 ± 2.9 g). However, this decrease in weight was restored in the herbal treatment groups (152.08 ± 0.702 g) (Fig. 4B). However, the herb-only group (146.81 ± 3.9 g) did not differ significantly from the negative control group. There was a significant ($P < 0.05$) increase in mean $\pm$ S.D. body temperature of rats in the infected group ($103.18 \pm 0.902^\circ\text{F}$) compared with the negative control ($98.03 \pm 0.44^\circ\text{F}$) (Fig. 4A). A sinusoidal or undulating pattern of increase on Day 1 post-infection, followed by a decrease on post-herbal treatment was observed in all treatment group rats. The mean $\pm$ S.D. body temperature of treated groups showed

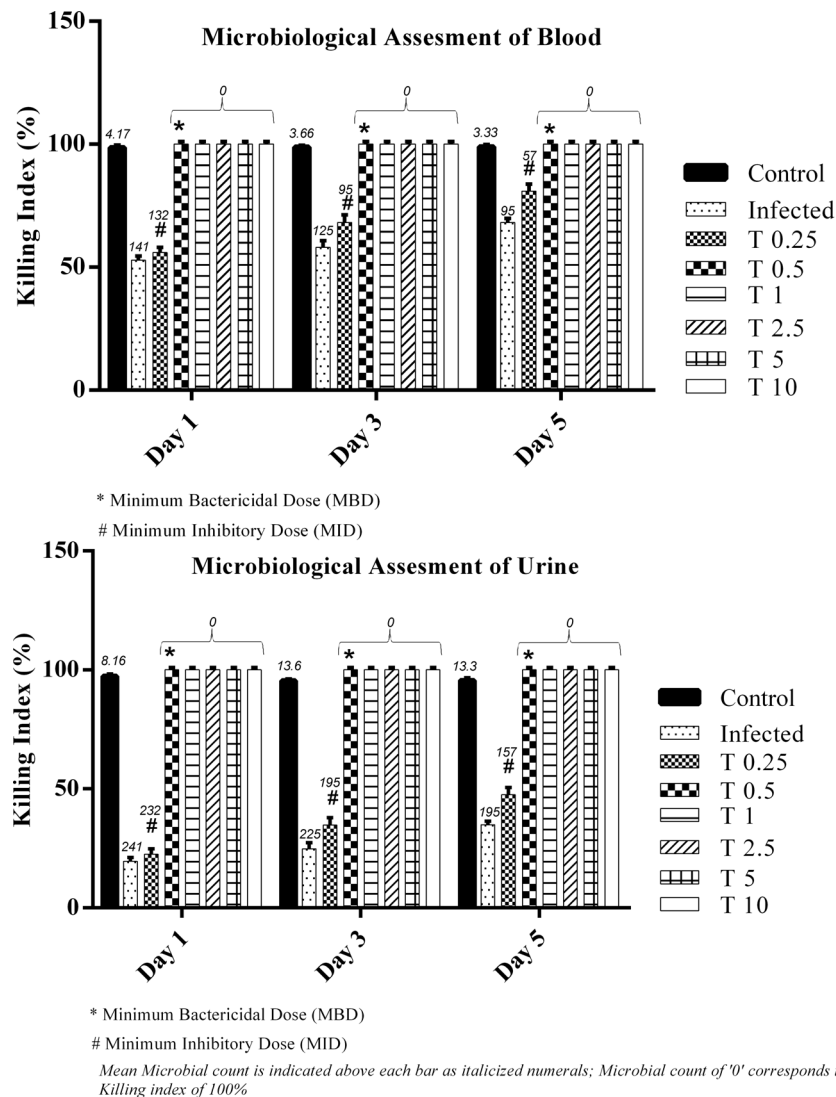


Fig. 3. In vivo antimicrobial activity of PTRC-2111-A. (A) Microbiological assessment of blood in negative control (infection only), infected and treatment (0.25–10 mg/kg BW) groups. The maximum reduction in microbial count (anti-CRE potential ca. $99.3 \pm 0.5\%$) was observed at 0.5 mg/kg BW (oral). (B) Microbiological assessment of urine in negative control, infected and treatment (0.25–10 mg/kg BW) groups, with 0.5 mg/kg BW (oral) being the most effective dose (minimum bactericidal dose). The killing index was calculated using the formula: killing index = (probable colonies – survival colonies) / (probable colonies). * Minimum bactericidal dose (MBD) represents the dose at which the microbial counts limits to zero; # minimum inhibitory dose (MID) represents the dose at which the microbial counts limits to 'too low to count' (TLTC). BW, body weight; CRE, carbapenem-resistant Enterobacteriaceae.

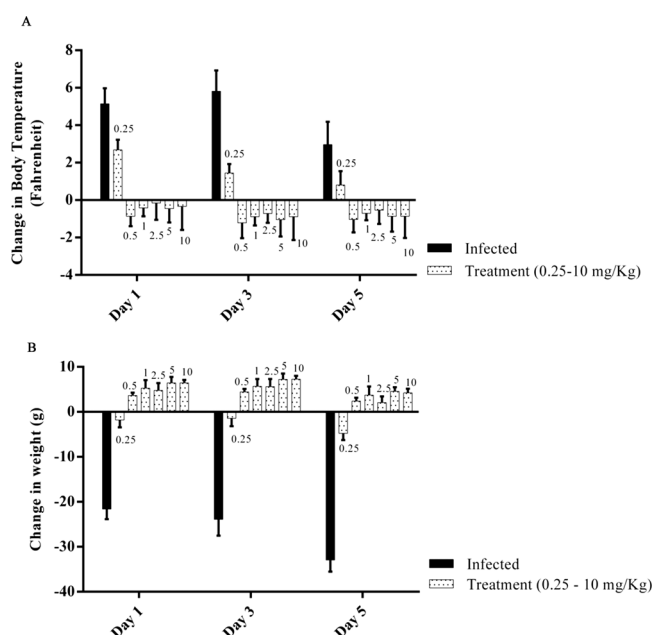


Fig. 4. Measurement of changes in physical parameters (body temperature and weight). (A) A sinusoidal or undulating pattern of increase in body temperature on Day 1 post-infection ($103.18 \pm 0.902^\circ\text{F}$), followed by a decrease post-herbal treatment was observed in all treatment group rats ($97.16 \pm 0.578^\circ\text{F}$). (B) A significant ($P < 0.05$) decrease in mean weight of all infected groups ($126.83 \pm 2.48\text{ g}$) was observed compared with the negative control group (infection only) ($148.43 \pm 2.95\text{ g}$). Such decrease in weight was restored in the herbal treatment groups at a dose $\geq 0.5\text{ mg/kg}$ body weight ($152.08 \pm 0.702\text{ g}$).

restoration to the normal range ($97.16 \pm 0.578^\circ\text{F}$). However, the herb only group ($98.28 \pm 0.172^\circ\text{F}$) did not significantly differ from the negative control group.

3.6. Haematology and biochemistry

Administration of PTRC-2111-A (0.5 mg/kg BW , oral) alone produced negligible aberrations in the haematology profile. However, there were significant changes in haematological parameters in the infected groups (Table 1). White blood cell, granulocyte (eosinophil) and agranulocyte (monocyte) counts of the infected group exhibited $50.8 \pm 0.05\%$, $19.06 \pm 0.04\%$ and $99.3 \pm 0.05\%$ relative increases, respectively, compared with the negative control group ($P = 0.001$). Furthermore, the red blood cell count, haemoglobin, Hct, PLT, RDW, MPV and PDW of the infected group showed a relative decrease in a range of $7.18\text{--}23.96\%$ ($P = 0.01$). Other haematological parameters, i.e. MCV, MCH, MCHC and lymphocyte count (LY), exhibited negligible changes in any of the groups [$F_{\text{MCV}}(3, 24) = 0.189$, $P = 0.902$], [$F_{\text{MCH}}(3, 24) = 0.014$, $P = 0.998$], [$F_{\text{MCHC}}(3, 24) = 0.014$, $P = 0.998$] and [$F_{\text{LY}}(3, 24) = 0.014$, $P = 0.998$].

It is evident from the serum biochemistry analysis that the extract and its interactions in the negative control and treatment groups exhibited non-significant differences in liver and kidney function tests, namely ALT, AST, LDH, GGT and ALP activities, whereas an increase in activities of liver and kidney enzymes indicated that there was significant ($P < 0.05$) liver and kidney dysfunction in the infected groups (Table 1).

Also, a high level of urea and creatinine in the blood of the infected group was indicative of severe kidney damage. TAS of blood of the infected group showed a significant ($P < 0.05$) reduction of nearly 60% compared with the negative control group, thereby indicating an indirect effect on innate immunity.

Table 1
Haematological and biochemical profile.

Evaluated parameter (mean $\pm$ SEM)	Reference standard	Control	Infected	Herbal treatment (0.5 mg/kg BW, oral)
WBC ($10^3/\mu\text{L}$)	6–16	9.2 ± 0.06	18.7 ± 0.05^a	11.4 ± 0.05
RBC ($10^6/\mu\text{L}$)	6–10	6.5 ± 0.1	5.16 ± 0.25^b	7.3 ± 0.02
Hb (g/dL)	11–19.2	11.2 ± 0.05	9.4 ± 0.3^b	12.6 ± 0.1
Hct (%)	34–54	35.0 ± 0.15	27.6 ± 0.01^b	37.2 ± 0.05
MCV (fL)	48–70	53.5 ± 0.03	53.5 ± 0.04^c	53.4 ± 0.01
MCH (pg)	11–19	17.1 ± 0.02	16.7 ± 0.05^c	18.2 ± 0.05
MCHC (g/dL)	30–40	32.0 ± 0.1	30.9 ± 0.05^c	34.1 ± 0.1
PLT ($10^3/\mu\text{L}$)	500–1300	703.0 ± 0.05	470 ± 0.5^b	834.6 ± 0.05
LY (%)	65–85	83.0 ± 0.1	83.7 ± 0.1^c	81.4 ± 0.02
MO (%)	0–5	0	3 ± 0.05^a	0
EO (%)	0–6	0	2 ± 0.05^a	0
GR (%)	12–18	17.4 ± 0.04	21.5 ± 0.05^a	15.3 ± 0.1
RDW (%)	10–13	12.1 ± 0.02	9.2 ± 0.01^b	11.9 ± 0.05
MPV (fL)	4–5	4.0 ± 0.05	3.1 ± 0.1^b	4.7 ± 0.1
PDW (%)	17–19	18.1 ± 0.04	16.8 ± 0.03^b	18.9 ± 0.05
ALT (IU/L)	35–80	43 ± 0.05	87 ± 0.05^a	47 ± 0.03
AST (IU/L)	1.56–100	89 ± 0.1	393 ± 0.5^a	89 ± 0.5
LDH (IU/L)	1000–1200	1010 ± 0.5	1359 ± 0.5^a	1034 ± 0.5
GGT (IU/L)	10–12	11.5 ± 0.01	13.7 ± 0.1^a	12 ± 0.05
ALP (IU/L)	150–250	159 ± 0.5	290 ± 0.5^a	168 ± 0.5
A/G ratio	0.65–1.8	0.72 ± 0.05	0.24 ± 0.02^b	0.90 ± 0.01
TAS (mmol/L)	0.5–2.5	0.75 ± 0.05	0.3 ± 0.03^b	1.5 ± 0.01
Urea (mmol/L)	4–6	5.35 ± 0.05	9.4 ± 0.05^a	4.66 ± 0.01
Creatinine ($\mu\text{mol/L}$)	50–57	53 ± 0.5	67 ± 0.5^a	53.9 ± 0.05

SEM, standard error of the mean; BW, body weight; WBC, white blood cells; RBC, red blood cells; Hb, haemoglobin; Hct, haematocrit/packed cell volume; MCV, mean corpuscular volume; MCH, mean corpuscular haemoglobin; MCHC, mean corpuscular haemoglobin concentration; PLT, platelet count; LY, lymphocytes; MO, monocytes; EO, eosinophils; GR, granulocytes; RDW, red blood cell distribution width; MPV, mean platelet volume; PDW, platelet distribution width; ALT, alanine aminotransferase; AST, aspartate aminotransferase; LDH, lactate dehydrogenase; GGT, γ -glutamyl transferase; ALP, alkaline phosphatase; A/G, albumin/globulin; TAS, total antioxidant status.

^a Significant increase in the estimated value compared with the reference standard and control group.

^b Significant decrease in the estimated value compared with the reference standard and control group.

^c Negligible change in the estimated value compared with the reference standard and control group.

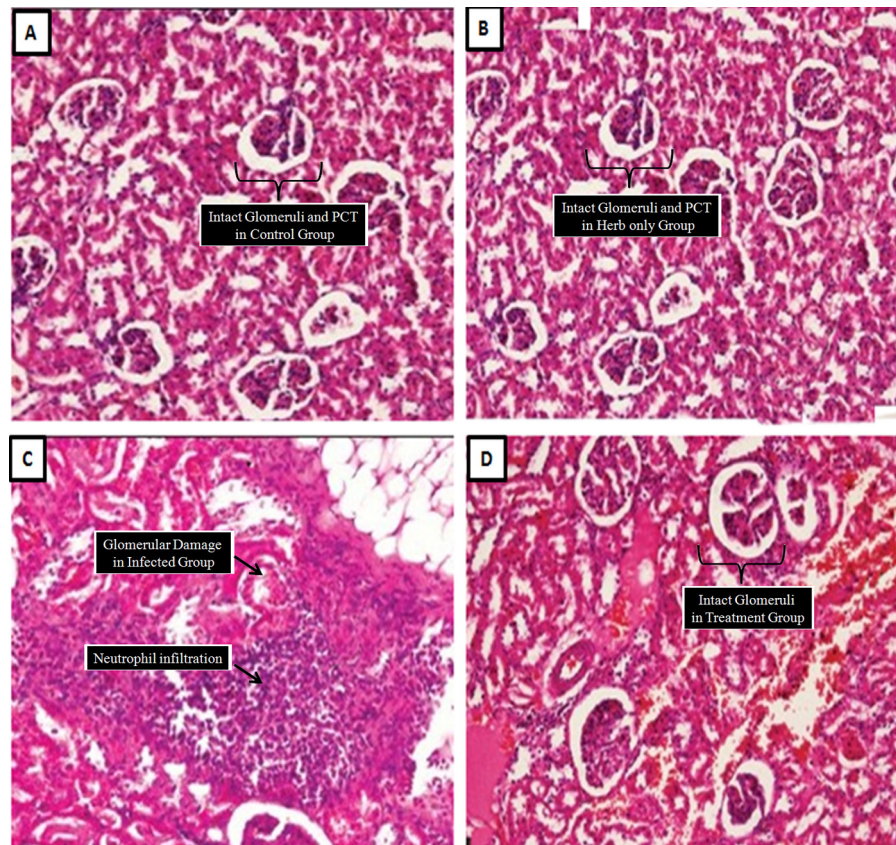


Fig. 5. Histopathological changes in the kidneys of control and experimental rats. (A) The negative control (infection only) rat kidney shows glomeruli and proximal convoluted tubules (PCT). (B) Kidney of rat treated with PTRC-2111-A, with no tissue damage. (C) Kidney of rat infected with NDM-1-producing *Escherichia coli*, with tubules showing neutrophil infiltration and glomerular damage. (D) Kidney of rat treated with PTRC-2111-A after *E. coli* infection, without any signs of glomerular damage, indicating restoration of kidney function.

Furthermore, there was a significant three-fold decrease in A/G ratio for infected groups compared with the negative control group.

3.7. Histopathological assessment

CRE-infected rats were morbid and showed signs of histological damage in the kidney. In addition, infected rats presented with symptoms of haemorrhage and glomerular damage (Fig. 5). There were multiple small necrotizing nodular lesions in the kidney parenchyma with a distorted proximal convoluted tube. Conversely, clinical signs and histological damage were rarely observed in rats in the uninoculated control, herbal only and treatment (0.5 mg/kg BW, oral) groups.

4. Discussion

Constantly re-emerging patterns of resistance among NDM-harboring isolates makes it imperative to monitor NDM-1 in order to understand its epidemiology, threat potential and varying virulence [10,28–31].

NDM-1-producing *E. coli* infects the host, ultimately causing peritonitis [32]. Peritonitis is an inflammatory condition of the peritoneum with physiological manifestations of increased capillary permeability and vasodilation that occurs secondary to microbial contamination [33]. In the present study, an analogous murine model of micro-organism-induced peritonitis in Sprague Dawley rats was used with an optimised intraperitoneal route of

CRE inoculation, a peak infection period of 24 h (Day 1) and herbal drug administration by the oral route.

Efficacy analysis showed that the maximum reduction in microbial count of CRE was observed at a single dosage of 0.5 mg/kg BW (oral) when administered prophylactically post-exposure (+24 h) (Fig. 3). Earlier studies confirmed the present findings, as *Berberis* sp. extract exhibited in vitro bactericidal activity against *V. cholerae* and *Staphylococcus aureus* at concentrations of 35 µg/mL [34] and 50 µg/mL [35], respectively. Berberine at a dose of 0.5 mg/kg BW (oral) has been reported to inhibit *Chlamydia trachomatis* in a chick embryo model [36]. Furthermore, *Berberis* sp. extract has also demonstrated activity at 6 mg/kg BW (oral) against *Streptococcus pneumoniae* in an in vivo study with white mice [37].

Sudden onset of fever and a significant reduction in weight are some of the physical symptoms associated with NDM-1-producing *E. coli* infection [25]. However, these physical symptoms might not be the definitive determinants of infection but clearly indicate inflammation and onset of bacterial infection. In the present study, there was a significant ($P < 0.05$) decrease in body weight and increase in body temperature of all infected groups compared with the negative control and herbal treatment groups (Fig. 4). These findings indicated the symptom-alleviating potential of the said compound.

CRE infection hampers several immunological and biochemical parameters of the host, thereby causing debilitating effects [38]. Haematological and biochemical profiling of the infected group (compared with the negative control) revealed a significant ($P < 0.05$) increase in immunological cell counts of total white blood cells (50.8% increase), eosinophils (19.06% increase) and

Table 2

Correlation analysis between pathophysiological indicators and microbial count (onset of disease) in a given population of Sprague Dawley rats.

Pathophysiological indicator	Microbial count (onset of disease)	
	<i>r</i>	<i>P</i>
Weight ^a	–0.626	0.001
Temperature ^b	0.898	0.001
WBC ^b	0.601	0.002
RBC ^a	–0.565	0.004
HGB ^a	–0.596	0.002
HCT ^a	–0.614	0.001
MCV ^c	0.093	0.665
MCH ^c	–0.020	0.927
MCHC ^c	–0.020	0.927
PLT ^a	–0.615	0.001
LY ^c	–0.403	0.843
MO ^b	0.603	0.002
EO ^b	0.599	0.002
GR ^b	0.590	0.002
RDW ^b	0.600	0.002
MPV ^a	–0.576	0.003
PDW ^a	–0.591	0.002
ALT ^b	0.612	0.001
AST ^b	0.616	0.001
LDH ^b	0.613	0.001
GGT ^b	0.599	0.002
ALP ^b	0.616	0.001
A/G ratio ^a	–0.598	0.002
TAS ^a	–0.540	0.006
Urea ^b	0.610	0.002
Creatinine ^b	0.618	0.001

WBC, white blood cells; RBC, red blood cells; Hb, haemoglobin; Hct, haematocrit/packed cell volume; MCV, mean corpuscular volume; MCH, mean corpuscular haemoglobin; MCHC, mean corpuscular haemoglobin concentration; PLT, platelet count; LY, lymphocytes; MO, monocytes; EO, eosinophils; GR, granulocytes; RDW, red blood cell distribution width; MPV, mean platelet volume; PDW, platelet distribution width; ALT, alanine aminotransferase; AST, aspartate aminotransferase; LDH, lactate dehydrogenase; GGT, γ -glutamyl transferase; ALP, alkaline phosphatase; A/G, albumin/globulin; TAS, total antioxidant status.

^a Negative correlation between the studied parameter and onset of disease as indicated by a significant decrease in the estimated value compared with the reference standard and control group.

^b Positive correlation between the studied parameter and onset of disease as indicated by a significant increase in the estimated value compared with the reference standard and control group.

^c Non-significant relationship between the studied parameters and onset of disease as indicated by negligible change in the estimated value compared with the reference standard and control group.

monocytes (99.3% increase). Such enhancement of immunological surveillance was directly linked to the CRE-mediated systemic infection, like colibacillosis and peritonitis, leading to leukocytosis, eosinophilia or monocytosis [39].

A positive correlation was found between the increase in immunological cell counts and infection ($r=0.601$, $n=24$, $P=0.002$), as shown in Table 2. Likewise, there was a direct correlation between the decrease in haematological indices and induction of disease ($r=0.596$, $n=24$, $P=0.002$). Such a decrease was an indirect indication of the destruction of erythrocytes by haemolytic enzymes produced by CRE [40].

Furthermore, a decrease in the number of platelets was attributed directly to CRE systemic infection, wherein the platelet surface membrane served as the site of antigen–antibody reactions, during which large numbers of platelets were destroyed, ultimately leading to idiopathic thrombocytopenia with decreased MPV and PDW [40]. Such a significant decrease was also found in direct correlation with onset of CRE infection cycle ($r=0.596$, $n=24$, $P=0.002$). The PTRC-2111-A-treated group showed absolute recovery of the haematological profile as well as restoration from idiopathic thrombocytopenia.

In view of the associated toxicity of last-line drugs and the impact of NDM-1 infection on baseline organ function, it was found essential to determine the hepatorenal protective ability of PTRC-2111-A. Liver function tests of the infected group showed significantly elevated levels of serum ALT ($r=0.612$), AST ($r=0.616$), LDH ($r=0.613$), GGT ($r=0.599$) and ALP ($r=0.616$). Such significant increases are indicative of the septicemic effect of CRE, which might lead to renal and hepatic failure if untreated. The renal dysfunction markers such as elevated levels of creatinine ($r=0.618$) and urea ($r=0.610$) also showed a similar trend. CRE infection further led to hypoalbuminemia as indicated by a reduced A/G ratio in the infected group. Such reduced A/G ratio ($r=-0.598$, $n=24$, $P=0.002$) was indicative of liver damage, which is the primary site of intrahepatic albumin synthesis [33]. The treated group showed complete protection against CRE-induced hepatorenal oxidative stress.

An important hallmark of the pathogenesis of CRE is the development of acute renal failure by inducing renal peritonitis and haemolytic–uraemic syndrome [34]. Our model presented the symptoms of haemorrhage and glomerular damage in agreement with previous reports of *E. coli*-induced renal peritonitis in rats [22]. Glomerular necrosis as observed in the case of the CRE-infected group could possibly be concomitant with an increase in plasma creatinine and urea concentrations as well as activation of oxidative damage and infiltration of macrophages into the glomeruli. However, no such renal damage was observed in the negative control and treated groups, thereby indicating the nephroprotective potential and symptomatic relief-providing capabilities of PTRC-2111-A (Fig. 5).

This study has also elucidated the antioxidative potential of PTRC-2111-A and its probable role in maintaining homeostasis. Nearly 60% decrease in the TAS of the CRE-infected group was observed compared with the negative control group, thereby indicating a positive correlation between decrease in TAS and CRE infection ($r=0.540$, $n=24$, $P=0.006$). Moreover, this indicated the destructive role of reactive oxygen species-induced oxidative stress in the above mentioned pre-clinical pathological manifestations. PTRC-2111-A restored the threshold TAS values with a single dose of 0.5 mg/gg BW (oral) prophylactically post-exposure.

LC–MS analysis of the extract PTRC-2111-A confirmed that the main components of the extract were isoquinoline alkaloids (Fig. 1). The most probable bactericidal mechanisms of these isoquinoline alkaloids could be bacterial DNA intercalation, inhibition of topoisomerases, inhibition of protein biosynthesis, causing mitochondrial fragmentation and induction of apoptosis in bacterial cells [35,36]. These studies hence provide cross-evidence for the pre-clinical efficacy of *B. aristata* in the management of CRE-infected patients.

5. Conclusion

Morbidity analysis in a peritonitis pre-clinical model revealed that PTRC-2111-A, an aquo-alcoholic standardised extract of *B. aristata*, was found to be therapeutically potent at a dose of 0.5 mg/kg BW (oral) against carbapenem-resistant *E. coli* infection.

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Ethical approval

Not required.

Competing interests

None declared.

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Appendix A. Supplementary data

Supplementary data associated with this article can be found, in the online version, at <http://dx.doi.org/10.1016/j.jgar.2016.12.018>.

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